ELECTRICAL ENGINEERING
UNIT 3: TECHNICAL TOOLS, CERTIFICATIONS & COMPETITIONS
Curriculum: Lovely Professional University | Department of Electrical Engineering
Reference: IEEE Standards Association / MATLAB & Simulink Power Systems Toolbox

>> 3.1 Industry-Standard Simulation Software Suites

Modern electrical engineering demands rigorous computational simulation prior to physical prototyping:
- MATLAB / Simulink & Simscape Electrical: Comprehensive numerical modeling of multi-domain physical systems. Modules include power systems, motor drives, semiconductor power losses, and control algorithm auto-code generation (Embedded Coder for TI DSPs).
- PSCAD / EMTDC: Transient electromagnetic analysis software. Utilized globally for lightning surge propagation, transformer core saturation, HVDC converter firing angle dynamics, and sub-synchronous resonance (SSR).
- ETAP (Electrical Transient Analyzer Program): The enterprise standard for industrial power networks. Key capabilities: Newton-Raphson load flow, balanced and unbalanced short circuit analysis (IEC 60909), protective relay coordination, and arc flash safety calculations.
- ANSYS Maxwell: Low-frequency electromagnetic Finite Element Analysis (FEA). Used to design magnetic circuits, optimize stator/rotor slot geometry in electric motors, minimize eddy currents, and calculate torque ripple.

>> 3.2 Hardware Prototyping & Digital Embedded Controllers

Real-time digital control bridges software algorithms with high-power hardware:
- Digital Signal Processors (DSP): Texas Instruments C2000 family (e.g., TMS320F28379D, F280049C) featuring high-resolution Pulse Width Modulation (HRPWM) with 150 picosecond resolution, dual-core architecture, and ultra-fast integrated SAR ADCs.
- Microcontrollers & SoC: STM32 (ARM Cortex-M4/M7 with floating point unit), ESP32 (dual-core 240 MHz for IoT-connected smart energy meters), and Microchip dsPIC.
- Hardware-in-the-Loop (HIL) Simulation: OPAL-RT and Typhoon HIL systems executing real-time FPGA-based solver models at 1-microsecond step times to test actual physical BMS and inverter controllers before high-voltage deployment.

>> 3.3 High-Power PCB Design & Electromagnetic Compatibility (EMC)

Power electronic hardware design requires strict adherence to physical layout standards:
- PCB Layout & Toolchains: Altium Designer and KiCad for multi-layer power PCB design.
- High-Current Trace Sizing: IPC-2152 standard calculations for trace width, copper thickness (2 oz, 3 oz, or 4 oz copper), and internal via thermal stitching.
- Creepage and Clearance Distances: IEC 60664-1 compliance to prevent dielectric

flashover across high-voltage traces under varying pollution degrees.
- Thermal Management: Direct Bonded Copper (DBC) ceramic substrates, insulated metal substrates (IMS), heat sink thermal resistance (R_th_ha) modeling, and forced-air fan CFM sizing.

>> 3.4 Professional Certifications & Global Engineering Standards

Professional credentials validate specialized technical competence:
- Certified Energy Auditor (CEA) & Energy Manager (CEM): Administered by the Bureau of Energy Efficiency (BEE), Ministry of Power, Government of India.
- Industrial Automation Certifications: Certified PLC/SCADA Systems Specialist (Siemens SITRAIN, Schneider Electric Automation Academy).
- Key Engineering Standards: IEEE 1547 (Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces), IEEE 519 (Harmonic Control in Electric Power Systems), and IEC 61850 (Communication Networks and Systems for Power Utility Automation).

DETAILED WORKED EXAMPLES & ENGINEERING COMPUTATIONS

>> Example 3.1: ETAP Power Flow and Shunt Capacitor Sizing

Statement: A 33 kV utility substation supplies power to a remote manufacturing cluster over an 8 km, 3-phase radial distribution feeder. Feeder parameters and load conditions are:
- Line Impedance: Z_line = (0.32 + j 0.58) Ohms per km
- Connected Industrial Load: S_load = 12 MVA at a lagging power factor of 0.80
- Substation Sending End Bus Voltage: V_s = 33.0 kV (line-to-line)
(a) Calculate the total feeder loop impedance, load current, and receiving end line-to-line bus voltage (V_r) without compensation.
(b) Calculate the feeder percentage voltage regulation and total active power transmission loss (kW).
(c) Determine the rating of a shunt capacitor bank (MVAR) required at the receiving bus to boost the power factor to 0.96 lagging and restore V_r to nominal 33 kV.

Step-by-Step Solution:
1. Feeder Total Impedance and Receiving Load Current:
- Total resistance: R_total = 0.32 * 8 = 2.56 Ohms
- Total inductive reactance: X_total = 0.58 * 8 = 4.64 Ohms
- Total feeder impedance: Z_line = (2.56 + j 4.64) Ohms (|Z| = sqrt(2.56^2 + 4.64^2) = 5.297 Ohms).
- Load active and reactive power:
P_load = S_load * cos(Phi) = 12 * 0.80 = 9.60 MW = 9,600 kW

Q_load = S_load * sin(Phi) = 12 * 0.60 = 7.20 MVAR = 7,200 kVAR
- Initial estimated line current: I_L = (12 * 10^6) / (sqrt(3) * 33,000) = 209.95 A.

2. Voltage Drop and Voltage Regulation Calculation:
- Approximate phase voltage drop formula: Delta_V_ph = I_L * (R * cos(Phi) + X * sin(Phi))
- Delta_V_ph = 209.95 * [(2.56 * 0.80) + (4.64 * 0.60)] = 209.95 * [2.048 + 2.784] = 209.95 * 4.832 = 1,014.5 V per phase.
- Line-to-line voltage drop: Delta_V_LL = sqrt(3) * 1,014.5 = 1,757.2 V = 1.76 kV.
- Receiving End Voltage: V_r = V_s - Delta_V_LL = 33.0 - 1.76 = 31.24 kV.
- Percentage Voltage Regulation = [(V_s - V_r) / V_r] * 100 = (1.76 / 31.24) * 100 = 5.63%.
- Total 3-Phase Line Loss: P_loss = 3 * I_L^2 * R_line = 3 * (209.95)^2 * 2.56 = 3 * 44,079 * 2.56 = 338,527 W = 338.53 kW.

3. Shunt Capacitor Bank Sizing for Power Factor Correction to 0.96 Lagging:
- Desired power factor: cos(Phi_new) = 0.96 => tan(Phi_new) = tan(acos(0.96)) = tan(16.26 deg) = 0.2917.
- Initial power factor angle: cos(Phi_old) = 0.80 => tan(Phi_old) = 0.7500.
- Required reactive power compensation: Q_cap = P_load * [tan(Phi_old) - tan(Phi_new)]
- Q_cap = 9.60 MW * [0.7500 - 0.2917] = 9.60 * 0.4583 = 4.40 MVAR.
- Practical Specification: Install a 4.5 MVAR, 33 kV automatic switched capacitor bank (three 1.5 MVAR stages).
- Post-compensation line current drops to: I_new = (9.60 * 10^6) / (sqrt(3) * 33,000 * 0.96) = 174.96 A.
- Reduced line losses: P_loss_new = 3 * (174.96)^2 * 2.56 = 235,093 W = 235.09 kW (Savings of 103.4 kW!).

>> Example 3.2: Digital PI Controller and Filter Sizing for Synchronous Buck Converter

Statement: An electrical engineer is designing a high-efficiency DC-DC synchronous buck converter to power an industrial IoT controller board from a 48 V telecom battery backup bus. The design requirements are:
- Input DC Voltage: V_in = 48.0 V (+/- 10%)
- Regulated Output DC Voltage: V_out = 12.0 V
- Rated Load Current: I_out = 15.0 A
- Switching Frequency: f_sw = 100 kHz (T_sw = 10 microseconds)
- Maximum Allowed Inductor Current Ripple: Delta_I_L <= 20% of rated I_out
- Maximum Allowed Output Voltage Ripple: Delta_V_out <= 1.0% of nominal V_out (120 mV)
(a) Calculate the steady-state duty cycle (D).
(b) Calculate the minimum filter inductance (L) and output filter capacitance (C).
(c) Determine the digital sampling frequency and discrete PI controller transfer function parameters using Bilinear Tustin Transformation for continuous gains K_p = 1.8 and K_i = 850.

Step-by-Step Solution:
1. Calculation of Duty Cycle (D):
- Neglecting diode and switch conduction drops: D = V_out / V_in = 12.0 / 48.0 = 0.25 (25.0%).

2. Sizing the Power Inductor (L):
- Permissible inductor ripple current: Delta_I_L = 0.20 * 15.0 A = 3.0 A.
- Inductor volt-second balance: V_L = L * (Delta_I_L / Delta_t) = (V_in - V_out)
- Since Delta_t = D * T_sw = D / f_sw:
L = [(V_in - V_out) * D] / (f_sw * Delta_I_L)
L = [(48.0 - 12.0) * 0.25] / (100,000 * 3.0) = [36.0 * 0.25] / 300,000 = 9.0 / 300,000 = 30.0 microHenries.
- Standard E-series selection: 33 microH, 20 A saturation current shielded power inductor.

3. Sizing the Output Filter Capacitor (C):
- In a buck converter, the inductor ripple current flows entirely into the output capacitor:
Delta_V_out = Delta_I_L / (8 * f_sw * C)
C = Delta_I_L / (8 * f_sw * Delta_V_out)
- Given Delta_V_out = 0.01 * 12.0 = 0.12 V:
C = 3.0 / (8 * 100,000 * 0.12) = 3.0 / 96,000 = 31.25 microFarads.
- Accounting for ceramic capacitor DC-bias capacitance degradation (40% loss at 12V DC):
C_actual = 31.25 / 0.60 = 52.08 microF. Select four 22 microF, 25V X7R ceramic capacitors in parallel (88 microF total).

4. Discrete PI Controller Discretization (Tustin Transform):
- Continuous PI controller: G_c(s) = K_p + (K_i / s)
- Sampling time T_s chosen equal to switching period T_sw = 10 microseconds (100 kHz sampling).
- Tustin substitution: s = (2 / T_s) * ((1 - z^-1) / (1 + z^-1))
- G_c(z) = K_p + K_i * (T_s / 2) * ((1 + z^-1) / (1 - z^-1))
- Numerical values: K_p = 1.8; K_i * (T_s / 2) = 850 * (10 * 10^-6 / 2) = 850 * 5 * 10^-6 = 0.00425.
- G_c(z) = [(1.8 + 0.00425) - (1.8 - 0.00425) * z^-1] / [1 - z^-1]
- Difference Equation for DSP implementation:
u[n] = u[n-1] + 1.80425 * e[n] - 1.79575 * e[n-1], where e[n] is the voltage error.

>> Example 3.3: Formula Student Electric EV Accumulator Pack Design

Statement: A student racing team participating in Formula Bharat Electric is designing a 400 V nominal lithium-ion accumulator container complying with FSAE electric vehicle regulations:
- Lithium-ion Cell: High-discharge 21700 cylindrical cells (Molicel P42A)
- Nominal cell voltage: V_cell = 3.60 V (Fully charged = 4.20 V, Cut-off = 3.00 V)

- Cell capacity: C_cell = 4.2 Ah
- Cell internal AC resistance (1 kHz): R_cell = 10.0 milliOhms
- Maximum continuous discharge rating: 45 A per cell
- Competition Rules Constraints: Maximum allowed tractive system voltage = 400.0 V DC; Max power limit = 80.0 kW.

(a) Determine the cell series-parallel configuration (Ns-Np) to achieve 400 V peak voltage and deliver 80 kW.
(b) Calculate the total internal resistance of the battery pack and total internal I^2*R heat dissipation at 80 kW power draw.
(c) Determine the cooling airflow rate (CFM) required to prevent cell temperatures from exceeding 50 deg C ambient during an endurance heat run.

Step-by-Step Solution:
1. Series and Parallel Cell Configuration:
- Maximum pack voltage at 100% state of charge: V_pack_max = N_s * 4.20 V <= 400.0 V.
- N_s = floor(400.0 / 4.20) = floor(95.23) = 95 cells in series.
- Nominal pack voltage: V_pack_nom = 95 * 3.60 V = 342.0 V.
- Maximum current drawn at 80 kW power limit (at nominal voltage): I_pack_max = P_limit / V_pack_nom = 80,000 W / 342.0 V = 233.92 A.
- Parallel cell requirement: N_p = ceil(I_pack_max / I_cell_max) = ceil(233.92 / 45) = ceil(5.20) = 6 cells in parallel.
- Accumulator Architecture: 95S6P configuration.
- Total number of cells: N_total = 95 * 6 = 570 cells.
- Total Pack Energy Capacity: E_pack = 95 * 3.60 V * (6 * 4.2 Ah) = 342 V * 25.2 Ah = 8,618.4 Wh = 8.62 kWh.

2. Pack Internal Resistance and Thermal Heat Generation:
- Resistance of one string of 95 series cells: R_string = 95 * 10.0 milliOhms = 950 milliOhms = 0.95 Ohms.
- Total pack internal resistance (6 strings in parallel): R_pack = R_string / 6 = 0.95 / 6 = 0.1583 Ohms (158.3 milliOhms).
- Internal I^2*R heat dissipation at peak 233.92 A load: Q_heat = I_pack_max^2 * R_pack = (233.92)^2 * 0.1583 = 54,718.5 * 0.1583 = 8,662 W = 8.66 kW of thermal energy!

3. Thermal Management and Cooling Airflow Calculation:
- Permissible air temperature rise: Delta_T = T_max_cell - T_ambient = 50 deg C - 35 deg C = 15 deg C (15 K).
- Specific heat capacity of air: c_p = 1,005 J / (kg * K); Air density: rho = 1.184 kg/m^3.
- Required mass flow rate of cooling air: m_dot = Q_heat / (c_p * Delta_T) m_dot = 8,662 / (1,005 * 15) = 8,662 / 15,075 = 0.5746 kg/s.
- Volumetric flow rate: V_dot = m_dot / rho = 0.5746 / 1.184 = 0.4853 m^3/s.
- Convert to CFM (Cubic Feet per Minute, 1 m^3/s = 2,118.88 CFM): Airflow = 0.4853 * 2,118.88 = 1,028.3 CFM.
- Engineering Selection: Install two 12V high-static-pressure brushless blowers rated at

550 CFM each.

>> Example 3.4: High-Voltage PCB Creepage & Clearance Sizing (IEC 60664-1)

Statement: A power electronics engineer is laying out the printed circuit board for an off-grid solar inverter operating with an internal DC bus voltage of 600 V DC and rated impulse withstand voltage of 4.0 kV.
- Environmental Conditions: Pollution Degree 2 (standard industrial laboratory environment)
- PCB Substrate Material: FR-4 fiberglass (Comparative Tracking Index CTI = 175 V to 400 V, Material Group IIIa)
- Insulation Requirement: Reinforced / Double Isolation between high-voltage DC bus and low-voltage microcontroller logic.
(a) Determine the minimum clearance distance (through air) according to IEC 60664-1.
(b) Determine the minimum creepage distance (along the PCB surface) for basic and reinforced insulation.
(c) If space constraints restrict creepage to 4.0 mm, design an isolation milling slot to satisfy safety requirements.

Step-by-Step Solution:
1. Minimum Clearance Distance Determination (IEC 60664-1 Table F.2):
- Clearance is determined by the impulse withstand voltage (peak voltage transient):
- For an impulse withstand rating of 4.0 kV in an inhomogeneous electric field (Case B):
Minimum Basic Clearance = 3.0 mm.
- For Reinforced Insulation, standard mandates stepping up to the next higher impulse withstand tier (6.0 kV):
Minimum Reinforced Clearance = 5.5 mm.

2. Minimum Creepage Distance Determination (IEC 60664-1 Table F.4):
- Operating working RMS/DC voltage = 600 V.
- Material Group IIIa (175 <= CTI < 400), Pollution Degree 2:
- Basic Creepage Distance for 630 V interpolated from table = 6.3 mm.
- Reinforced Insulation requires doubling the basic creepage distance:
Reinforced Creepage = 2 * 6.3 mm = 12.6 mm.

3. Engineering Design of PCB Isolation Milling Slot:
- When linear PCB trace spacing cannot accommodate 12.6 mm creepage due to compact form factors:
- An air slot (isolation trench) is routed through the FR-4 board beneath the optocouplers/digital isolators.
- According to IEC 60664-1, an air slot having width >= 1.0 mm breaks the conductive tracking path along the surface.
- The creepage measurement path then traverses down through the slot boundary, through air, and up the other side.
- Path length = Surface distance to edge (4 mm) + Slot depth (1.6 mm board thickness) +

Slot bottom width (1.5 mm) + Board thickness (1.6 mm) + Surface distance (4 mm) = 12.7 mm.

- Conclusion: Routing a 1.5 mm wide milled slot provides 12.7 mm creepage, fully satisfying the 12.6 mm reinforced safety requirement!